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1 الملف: Structured Cabling System (SCS)

Technical Specification & Implementation Guide

Project: Soho Hills Tower A — Project 120

Location: Dubai Hills, Dubai, UAE

Consultant: National Engineering Bureau (NEB)

Main Contractor: New System Engineering LLC

MEP Contractor: SPOTS ELECTROMECHANICAL WORKS LLC

SECTION 1: INTRODUCTION

1.1 Purpose

This document specifies the technical requirements, materials, installation methods, testing procedures, and quality standards for the Structured Cabling System (SCS) to be installed at Project 120 — Soho Hills Tower A, Dubai Hills, Dubai, UAE.

1.2 Scope of Work

The scope includes the supply, installation, termination, testing, labeling, and commissioning of a complete end-to-end structured cabling system comprising:

- Fiber optic backbone cables (Single-mode OS2)
- UTP Cat 6 horizontal copper cables
- Fiber termination boxes (4-port)
- Consolidation cabinets (12U)
- Main Telecom Room (MTR) equipment
- GPON splitters (2x32)
- Patch panels (Copper & Fiber)
- All associated containment, pathways, and accessories

1.3 Applicable Standards

The installation shall comply with the following standards:

Standard	Description
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ISO/IEC 11801	Generic Cabling for Customer Premises
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EN 50173-1	Information Technology — Generic Cabling
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ANSI/TIA-568-C.2	Commercial Building Telecommunications Cabling
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ANSI/TIA-569	Commercial Building Standard for Pathways and Spaces
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TIA/EIA TSB-72	Centralized Optical Fiber Cabling Guidelines
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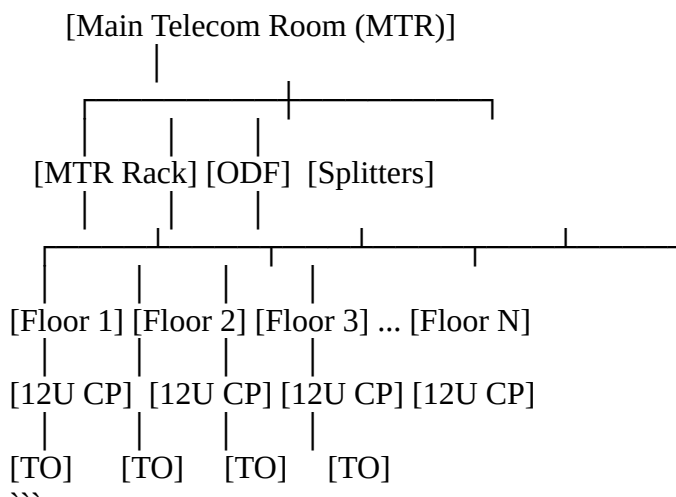
TIA/EIA TSB-75	Additional Horizontal Cabling Practices
IEC 60364-1	Electrical Installations of Buildings
IEC 60950	Safety of Information Technology Equipment
TRA / TDRA	UAE Telecommunications Regulatory Authority Requirements
Etisalat / du	UAE Telecom Operators Guidelines

SECTION 2: SYSTEM ARCHITECTURE

2.1 Network Topology

The structured cabling system shall be based on a star topology with the following hierarchy:

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2.2 System Components

A. Main Telecom Room (MTR)

- Located on Ground Floor or Basement
- Minimum dimensions: 3m × 3m × 3m
- Contains: 42U Floor Standing Rack, ODF, GPON Splitters, UPS

B. Floor Distributor (FD)

- Located on each floor
- Contains: 12U Wall Mount Cabinet
- Serves up to 3 floors per Mini ODF

C. Consolidation Point (CP)

- Located inside each apartment
- 12U Home Consolidation Cabinet
- Contains: Fiber Terminal Box, Copper Patch Panel, Power Socket

D. Telecommunications Outlet (TO)

- Located in each room
- Dual RJ45 Outlet with shutter

- Single-mode Fiber Outlet (optional)

SECTION 3: MATERIAL SPECIFICATIONS

3.1 Cat 6 UTP Copper Cable

Parameter	Specification
Category	Cat 6
Conductor	Solid Bare Copper
Conductor Size	23 AWG
Insulation	HDPE
Pairs	4 Twisted Pairs
Jacket Material	LSZH (Low Smoke Zero Halogen)
Jacket Color	Grey (GY)
External Diameter	6.3 ± 0.3 mm
Frequency	Up to 250 MHz
Impedance	$100 \pm 15 \Omega$ (1-100 MHz)
DC Resistance	$\leq 9.5 \Omega/100m$
Mutual Capacitance	< 55 pF/m
Propagation Delay Skew	< 20 ns/100m
Part Number	C06UI3-GY

Performance Standards:

- ISO/IEC 11801
- TIA-568-C.2
- EN 50288-6-1:2013
- IEC 60332-1-2 (Flame Retardant)
- IEC 61034-1/2 (Smoke Density)
- IEC 60754-2 (Halogen Content)

3.2 Fiber Optic Cable (Single-Mode OS2)

A. Backbone Fiber Cable (48-Core)

Parameter	Specification
Fiber Type	Single-Mode OS2
Standard	ITU-T G.657.A1/A2
Backward Compatible	ITU-T G.652.D
Core Diameter	9 μ m
Cladding Diameter	125 ± 1.0 μ m
Coating Diameter	245 ± 10 μ m
Attenuation @1310nm	≤ 0.4 dB/km
Attenuation @1550nm	≤ 0.3 dB/km
Jacket Material	LSZH
Fiber Count	48 cores

| Part Number | FCOS2-48-MTBI-3 |

B. Drop Fiber Cable (4-Core)

Parameter	Specification
Fiber Type	Single-Mode OS2
Fiber Count	4 cores
Cable Type	FTTH Drop Cable
Cross Section	Flat
Strength Member	FRP (Fiberglass Reinforced Plastic)
Jacket Material	LSZH
Bend Radius	10× Cable Diameter
Part Number	FCOS2-04-DRPI-3

3.3 Fiber Connectors

A. SC/APC Connector

Parameter	Specification
Type	SC/APC (Angled Physical Contact)
Insertion Loss	< 0.10 dB
Return Loss	≥ 60 dB
Durability	1000 mating cycles
Operating Temp	-20°C to +75°C
Standard	IEC 61754
Part Number	A-SCAS-C

B. LC/APC Connector

Parameter	Specification
Type	LC/APC
Insertion Loss	< 0.10 dB
Return Loss	≥ 60 dB
Durability	1000 mating cycles
Part Number	A-LCAD-C

3.4 Copper Connectors

Cat 6 Keystone Jack

Parameter	Specification
Category	Cat 6
Type	Unshielded RJ45
Angle	180°
Wire Gauge	AWG 22-26

Insertion Life	750 cycles minimum
Dielectric Strength	AC 750V / DC 1000V
Contact Material	Phosphor Bronze
Standard	ISO/IEC 11801, TIA/EIA 568C
Part Number	M06U180

3.5 Patch Panels

A. Copper Patch Panel (24-Port)

Parameter	Specification
Ports	24
Type	Unshielded
Size	19" 1U
Cable Management	Brush Type
Part Number	PP06U24-UL

B. Fiber Patch Panel (48-Port)

Parameter	Specification
Ports	48
Type	SC Simplex / LC Duplex
Size	19" 2U
Splice Cassette	48-way
Part Number	E-P-48-SC

3.6 Cabinets

A. Floor Standing Cabinet (42U)

Parameter	Specification
Height	42U
Width	800 mm
Depth	800 mm
Front Door	Glass Single Door
Rear Door	Steel Sheet Single Door
Side Panels	Included
Mounting	19" Angles
Castors	Yes
Fan Tray	4-Way on Top
Part Number	FS428811-9004

B. Wall Mount Cabinet (12U)

Parameter	Specification

Height	12U
Width	600 mm
Depth	150 mm (Flush) / 300 mm (Wall)
Front Door	Glass Door
Color	RAL 7035
Part Number	WMONUF12615-7035

3.7 GPON Splitters

Parameter	Specification
Type	Planar Lightwave Circuit (PLC)
Configuration	2×32
Connector Type	SC/APC (Etisalat) / LC/APC (du)
Operating Wavelength	1260 - 1635 nm
Insertion Loss	≤ 18 dB
Operating Temp	-24°C to +85°C
Standard	Telcordia GR-1209, GR-1221
Part Number	SPSCRM2321.5 (Etisalat) / SPLCRM2321.5 (du)

SECTION 4: INSTALLATION PROCEDURES

4.1 Pre-Installation

A. Site Survey

1. Verify all pathways and containment are installed per approved drawings
2. Check that all cable trays, conduits, and pull boxes are in place
3. Confirm MTR, FTR, and CP locations
4. Verify electrical supply (13A twin socket) in each CP

B. Material Inspection

1. Check all materials against the Approved Material Submittal
2. Verify cable reels for damage and correct lengths
3. Check continuity of fiber optic cables (Pre-installation testing)
4. Verify all connectors and patch panels

4.2 Copper Cable Installation

A. Cable Pulling

1. Verify cable pathway is free from sharp bends
2. Use cable rollers at corners
3. Maintain minimum bend radius: 4× cable diameter (25 mm)
4. Maximum pulling tension: 110 N (25 lbf)
5. Do not exceed 90 meters from patch panel to outlet

B. Cable Support

1. Use cable trays or baskets for horizontal runs

2. Maximum 60 cables per 100×50 mm tray
3. Support cables every 1.5 meters
4. Use Velcro ties (not plastic zip ties)

C. Cable Separation

1. Maintain minimum 300 mm separation from power cables
2. Cross power cables at 90° angle
3. Use separate conduits for telecom and power

D. Cable Labeling

1. Label both ends of each cable
2. Use unique identifier per approved labeling scheme
3. Machine-printed labels only

4.3 Fiber Optic Cable Installation

A. Pre-Installation Testing

1. Test each fiber reel with OTDR
2. Verify attenuation < 0.4 dB/km @1310nm
3. Document results

B. Cable Pulling

1. Maximum pulling tension: 100 N (short-term)
2. Minimum bend radius: 20× cable diameter (during installation)
3. Minimum bend radius: 10× cable diameter (after installation)
4. Use fiber optic lubricant if needed

C. Fiber Splicing

1. Use fusion splicer for all splices
2. Splice loss < 0.1 dB
3. Protect splices with splice protectors
4. Store in splice trays

D. Fiber Termination

1. Terminate in 4-port fiber terminal box
2. Use SC/APC pigtails for Etisalat
3. Use LC/APC pigtails for du
4. Clean all connectors before mating

4.4 Cabinet Installation

A. Floor Standing Cabinet

1. Install on raised floor or concrete base
2. Level and secure with anchor bolts
3. Install cable management (brush panels)
4. Install PDU (6-way horizontal)
5. Connect earth bonding

B. Wall Mount Cabinet

1. Install at height 1200 mm from finished floor
2. Secure to wall with anchor bolts
3. Ensure front door opens fully
4. Connect earth bonding

C. Home Consolidation Cabinet

1. Install inside each apartment near entrance
2. Height: 1200 mm from FFL
3. Do not install in kitchen, laundry, or bedroom
4. Ensure 13A power socket inside cabinet
5. Lockable front door

SECTION 5: TESTING & COMMISSIONING

5.1 Copper Cable Testing

A. Test Equipment

- Fluke DSX-5000 CableAnalyzer (Level II Tester)
- Calibrated per ISO/IEC 11801

B. Test Parameters

1. Wire Map
2. Length
3. Insertion Loss (Attenuation)
4. Return Loss
5. NEXT (Near-End Crosstalk)
6. PSNEXT (Power Sum NEXT)
7. ELFEXT (Equal Level Far-End Crosstalk)
8. PSELFEXT
9. ACR (Attenuation to Crosstalk Ratio)
10. PSACR
11. Delay Skew
12. Impedance
13. DC Loop Resistance

C. Pass/Fail Criteria

- All tests must PASS per Cat 6 limits
- 100% of installed links tested
- Failed links must be repaired and retested

D. Documentation

- Save results in Fluke LinkWare software
- Export to PDF for submission
- Submit to consultant for approval

5.2 Fiber Optic Testing

A. Test Equipment

- OTDR (Optical Time Domain Reflectometer)
- Light Source & Power Meter

B. Test Parameters

1. Continuity
2. Length
3. Attenuation @ 1310nm and 1550nm
4. Splice Loss
5. Connector Loss
6. Return Loss

C. Test Method

- Light Source + Power Meter: One direction (sufficient)
- OTDR: Both directions (required)
- Test 100% of installed fibers

D. Pass/Fail Criteria

- Attenuation $\leq$ design loss budget
- Splice Loss $\leq$ 0.1 dB
- Connector Loss $\leq$ 0.3 dB

5.3 System Commissioning

A. Visual Inspection

1. Verify all outlets are labeled
2. Verify all cables are dressed properly
3. Verify all connectors are clean
4. Verify earth bonding is complete

B. Functional Testing

1. Connect patch cords and verify link
2. Test IP connectivity (if applicable)
3. Verify GPON splitter operation

C. Documentation

1. As-Built Drawings
2. Test Reports (Copper + Fiber)
3. Labeling Schedule
4. Warranty Certificate

SECTION 6: QUALITY ASSURANCE

6.1 Quality Control

Activity	Responsibility	Frequency
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Material Inspection	QA/QC Engineer	Every delivery
Cable Pulling Inspection	Site Engineer	Daily
Termination Inspection	QA/QC Engineer	Per floor
Testing	QA/QC Engineer	100% links
Final Inspection	Consultant	Per milestone

6.2 Non-Conformance

Any non-conformance shall be:

1. Documented in NCR (Non-Conformance Report)
2. Corrected within 7 days
3. Retested and approved

6.3 Warranty

Item	Warranty Period
Product Warranty	1 year from TOC
Performance Warranty	25 years from TOC
Installation Warranty	1 year from TOC

SECTION 7: SAFETY REQUIREMENTS

7.1 Personal Protective Equipment (PPE)

- Safety Helmet
- Safety Shoes
- Safety Glasses
- Fluorescent Vest
- Safety Gloves
- Dust Mask (when required)

7.2 Working at Heights

1. Use scaffolding or ladders (approved)
2. Do not work alone
3. Use fall protection if > 2m
4. Barricade work area

7.3 Electrical Safety

1. Use LOTO (Lock-Out Tag-Out) procedure
2. Verify no live circuits
3. Use insulated tools
4. Follow PTW (Permit to Work) system

7.4 Emergency Procedures

1. First Aid training for all technicians
2. CPR training
3. Emergency contact list posted

4. Fire extinguishers available

SECTION 8: PROJECT SCHEDULE

Phase	Duration	Milestone
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Mobilization	1 week	Site setup
Containment Installation	4 weeks	Cable trays installed
Copper Cable Pulling	6 weeks	All Cat 6 installed
Fiber Cable Pulling	4 weeks	All fiber installed
Termination	4 weeks	All connectors terminated
Testing	3 weeks	100% links tested
Commissioning	1 week	System handed over
Total	23 weeks	

SECTION 9: DOCUMENTATION

9.1 Submittals

1. Material Submittal (Approved)
2. Shop Drawings (Approved)
3. Method Statement (Approved)
4. Testing & Commissioning Statement

9.2 As-Built Documentation

1. As-Built Drawings (CAD)
2. Cable Schedule (Excel)
3. Test Reports (PDF)
4. Labeling Schedule
5. Warranty Certificates
6. Operation & Maintenance Manual

SECTION 10: APPENDICES

Appendix A: Cable Schedule (Example)

Cable ID	From	To	Type	Length
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C-001	MTR-ODF-1	FTR-1F-MiniODF	48C SM	85m
C-002	FTR-1F-MiniODF	CP-101	4C SM	45m
C-003	FTR-1F-MiniODF	CP-102	4C SM	52m
D-001	CP-101	TO-101-A	Cat 6 UTP	25m
D-002	CP-101	TO-101-B	Cat 6 UTP	30m

Appendix B: Labeling Scheme

Cable Labels:

[Project Code]-[Floor]-[Type]-[Number]

Example: 120-1F-C-001

Outlet Labels:

[Flat Number]-[Room]-[Port]

Example: 101-BED1-A

Appendix C: Test Report Format

Link ID	Length	NEXT	Attenuation	Return Loss	Result
101-A	25.3m	45.2 dB	18.5 dB	22.1 dB	PASS
101-B	30.1m	43.8 dB	19.2 dB	21.5 dB	PASS

END OF DOCUMENT

Document Reference: SPOTS-120-SCS-SPEC-001

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Prepared by: SPOTS ELECTROMECHANICAL WORKS LLC